

Autonomous Data Intelligence Report

Dataset Summary

Before Cleaning

Rows: 891
Columns: 12
Missing Values: 866
Duplicate Rows: 0

After Cleaning

Rows: 891
Columns: 12
Missing Values: 0
Duplicate Rows: 0

Problem Detection

Target Column: Survived
Problem Type: classification

Model Performance

Logistic Regression: 0.8212
Random Forest: 0.8212
XGBoost: 0.8268
Best Model: XGBoost

Analytics Summary

Analytics insights temporarily unavailable.

AI Dataset Insights

Executive Summary Report: Titanic Survival Prediction --- ### 1. Dataset Summary * **Dataset Context**: This is the classic Titanic passenger dataset, used to predict passenger survival based on demographic, socio-economic, and travel characteristics. * **Target Variable**: `Survived` (Binary Classification: `0` = Deceased, `1` = Survived). * **Dataset Dimensions**: 891 rows and 12 columns. * **Feature Types**: * **Numerical Columns (7)**: `PassengerId`, `Survived` (Target), `Pclass` (Socio-economic class), `Age`, `SibSp` (Siblings/Spouses aboard), `Parch` (Parents/Children aboard), and `Fare`. * **Categorical Columns (5)**: `Name`, `Sex`, `Ticket`, `Cabin`, and `Embarked` (Port of embarkation). * **Data Volume**: Extremely lightweight, with a memory footprint of just **0.12 MB**. --- ### 2. Data Cleaning Impact * **Initial State**: The original dataset contained **866 missing values** (representing an overall **8.1% missingness** across the

data grid). In the Titanic dataset, missing values are historically concentrated in `Cabin` (high missingness), `Age` (moderate missingness), and `Embarked` (minimal missingness). * **Cleaning Process**: * A robust imputation strategy was successfully executed, bringing the missing value count down to **0**. * Crucially, **0** rows were dropped during the cleaning process (the row count remained steady at 891). This means missing values were intelligently imputed (e.g., using median values for `Age`, mode for `Embarked`, or a placeholder category like "Unknown" for `Cabin`), preserving 100% of the training instances. * No duplicate rows were detected, ensuring no synthetic bias was introduced. --- **3. Important Patterns & Domain Insights** Given the features in this dataset, several strong patterns typically dictate survival classification: * **The "Women and Children First" Protocol**: The `Sex` and `Age` columns are highly predictive. Female passengers and younger children historically have drastically higher survival rates. * **Socio-Economic Stratification**: The `Pclass` (1st, 2nd, or 3rd class) and `Fare` features act as strong proxies for safety. First-class passengers paying premium fares had preferential access to lifeboats. * **Family Size Impact**: Combining `SibSp` and `Parch` helps identify whether a passenger traveled alone or in a group. Large families or solo travelers historically faced lower survival rates compared to small, cohesive family units (size 2 to 4). --- **4. Model Evaluation** Three machine learning algorithms were trained and evaluated on the cleaned dataset: | Model | Accuracy / Performance Score | Status | | :--- | :--- | :--- | | **XGBoost** | **82.68%** | **Best Model (Selected)** | | **Random Forest** | 82.12% | Baseline Competitor | | **Logistic Regression** | 82.12% | Baseline Competitor | **Key Takeaways**: * **Top Performer**: **XGBoost** yielded the highest accuracy of **82.68%**. Its gradient boosting architecture excelled at capturing complex, non-linear interactions between variables (such as the interaction between being a young male in 3rd class vs. an older female in 1st class). * **Alternative Models**: Both Random Forest and Logistic Regression performed respectably at **82.12%**, indicating that the clean dataset holds strong linear relationships alongside non-linear ones. * **Artifact Deployment**: The best-performing model has been successfully serialized and saved to `ml\_engine/models/XGBoost.pkl`, ready for production inference. --- **5. Recommendations** To further improve model performance and deployment reliability, the following steps are recommended: 1. **Feature Engineering**: * **Title Extraction**: Extract titles (e.g., *Mr.*, *Mrs.*, *Miss*, *Master*) from the `Name` column. This acts as a highly predictive proxy for both age and social status. * **Family Size**: Combine `SibSp` and `Parch` into a single `Family\_Size` feature, and bin them into categories: `Solo`, `Small\_Family`, and `Large\_Family`. * **Deck Extraction**: Extract the first letter from the `Cabin` column to create a `Deck` feature (A, B, C, etc.), which correlates directly with structural safety and proximity to the top deck. 2. **Hyperparameter Tuning**: Run a randomized or grid search cross-validation specifically on the **XGBoost** model to tune parameters like `max\_depth`, `learning\_rate`, and `n\_estimators` to push accuracy beyond the 83% threshold. 3. **Model Explainability**: Utilize **SHAP** (SHapley Additive exPlanations) on the saved `XGBoost.pkl` model. This will visually prove to stakeholders exactly how features like `Sex`, `Pclass`, and `Age` influenced individual survival predictions.

Model Selection Reasoning

Model explanation unavailable: 429 You exceeded your current quota, please check your plan and billing details. For more information on this error, head to:

<https://ai.google.dev/gemini-api/docs/rate-limits>. To monitor your current usage, head to:

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